Penalties Sized to the Gain

When do random audits stop AI agents from over-using a shared resource?

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PhD catch-up
King's College London, 13 October 2026

The Problem

AI agents share limited things

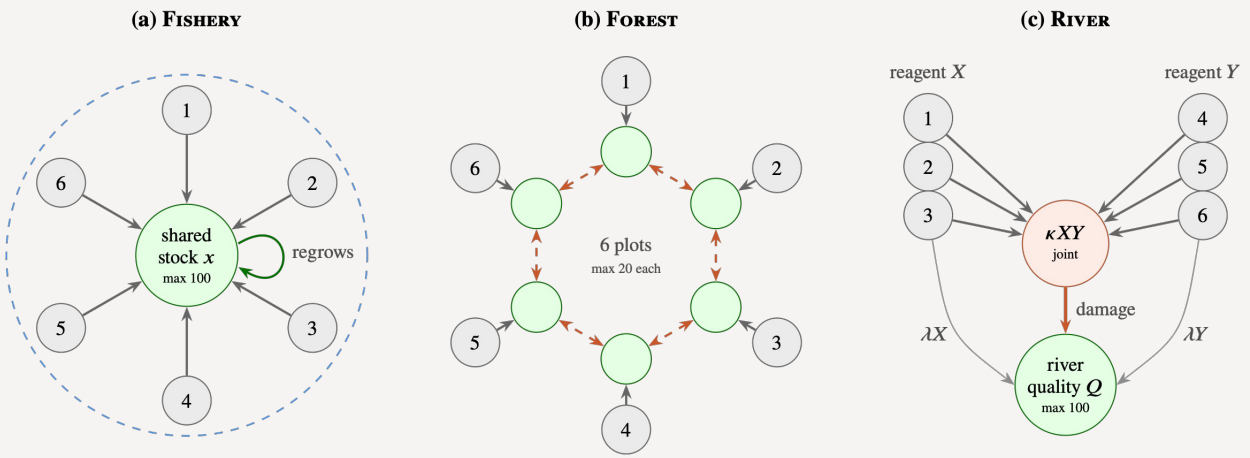
Compute, energy, money, a fish stock. Each agent is told how much it may take.

Checking is expensive

An overseer can only spot-check a few agents each round, like a tax audit or a fishing inspection.

What we don't know

Do spot-checks stop agents taking more than their share? And how big must the penalty be?



Three toy worlds with six agents each. Fishery: one shared stock. Forest: over-cutting damages your two neighbours. River: two pollutants that are harmful mainly together.

Three Questions

01

What must follow a catch?

A fine, a memory of the offence, or nothing: which of these makes spot-checks reduce harm?

02

Can we predict the stopping penalty?

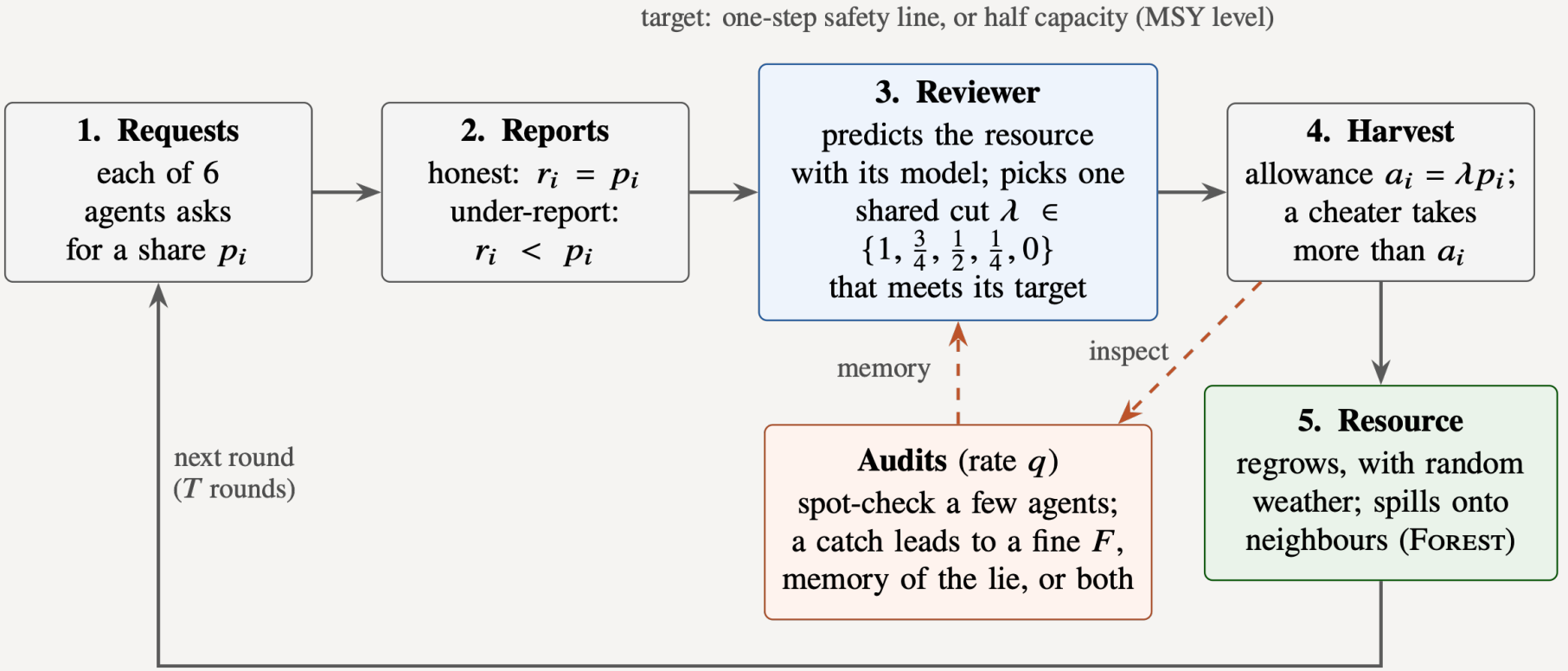
Economics says cheating stops once the expected fine beats the gain. Does that hold, and can we predict it in advance?

03

Do LLMs follow the same rule?

If four of the six agents are LLMs, do they stop where we predict, and which gain do they act on?

How One Round Works



Each round: agents ask for a share, a reviewer scales every request by one factor to keep the resource at half capacity, agents take their allowance or more, the resource regrows, and a few agents are spot-checked.

Harm = the resource falling below half capacity, where it regrows fastest

The Rule From Economics

$q \times F$

expected fine = chance of being checked × size of the fine

$1/6 \times 30$ t

= 5 t expected fine per round of cheating

≈ 4.5 t

what a model gained per round of cheating, with no fine

Cheat while the gain beats the expected fine

This is Becker's 1968 rule. So a 30 t fine should stop an agent that gains 4.5 t per round, and a 24 t fine (4 t expected) should not.

But which gain?

The gain from one round, or the gain over the whole game once the stock is depleted? The two differ, and that turns out to matter.

Finding 1: Catching Alone Barely Helps

Remembering each caught lie works

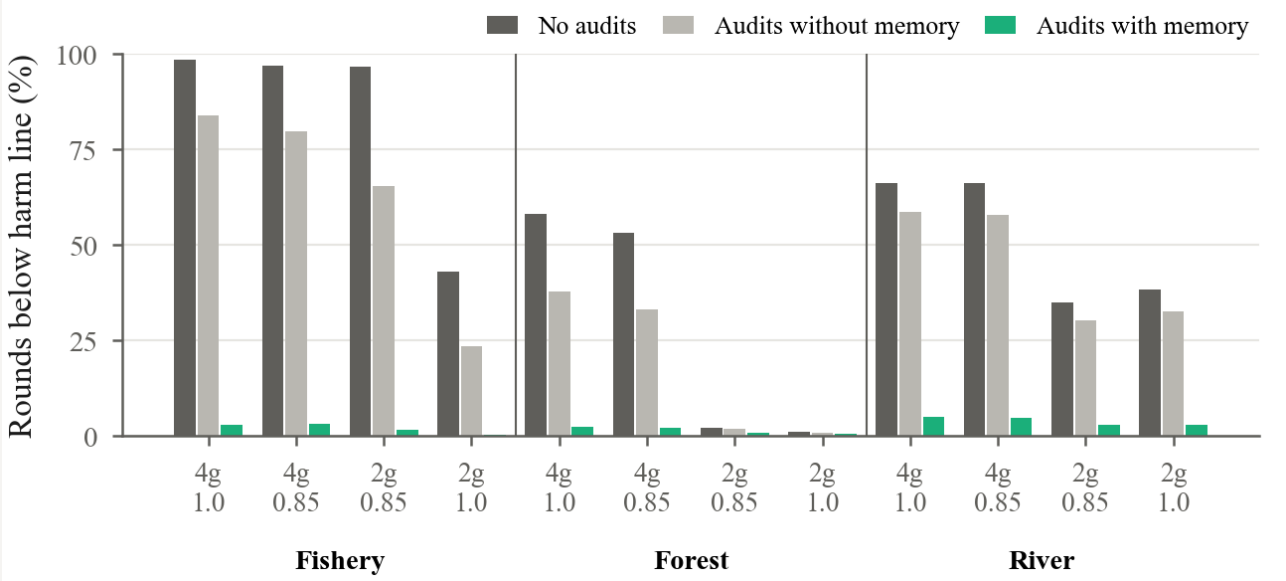
Memory gave less harm than audits without memory in 12 of 12 settings across the three games.

The numbers

Rounds below half capacity fell from 43–99% to 0.3–3% in Fishery and from 35–66% to 3–5% in River.

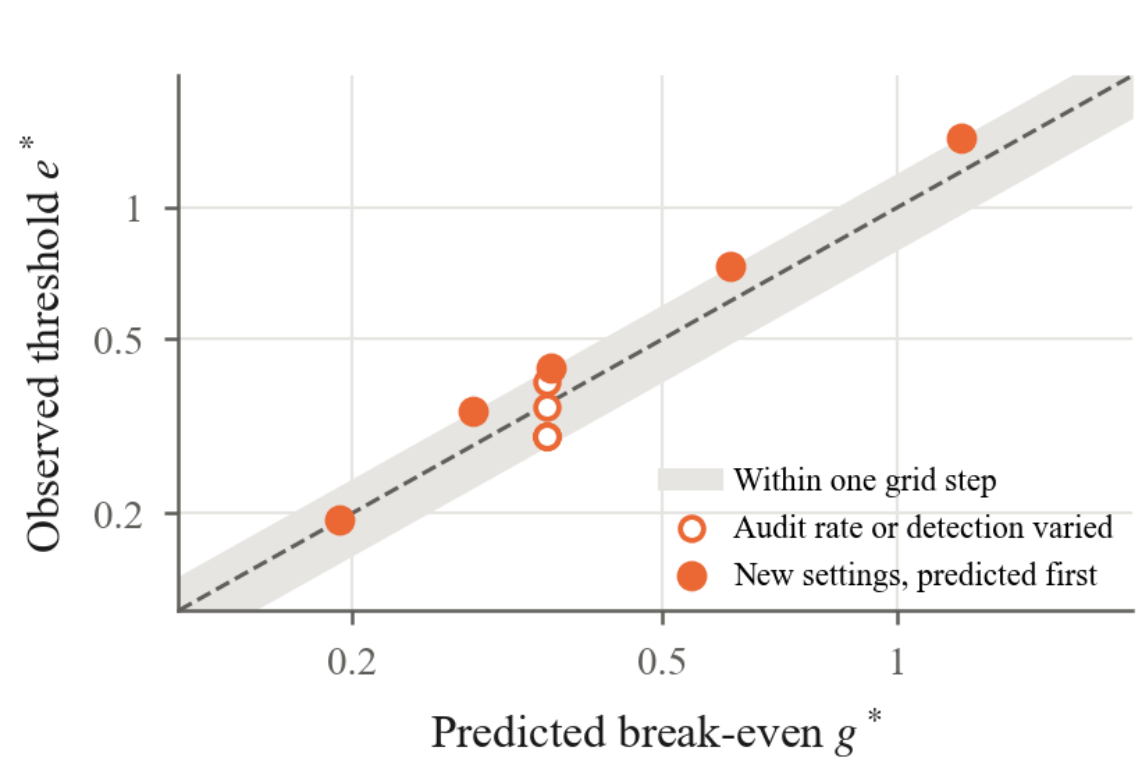
Audits without memory

Correcting only the current round removed just 10–46% of the harm.



Share of rounds below half capacity with lying agents. Dark: no audits. Light grey: audits without memory. Green: audits with memory.

Finding 2: A Predictable Threshold



Observed against predicted threshold. Filled: new settings predicted first. Open: audit rate or detection varied.

Predicted before testing

From runs without audits, we predicted where cheating would stop in five new settings.

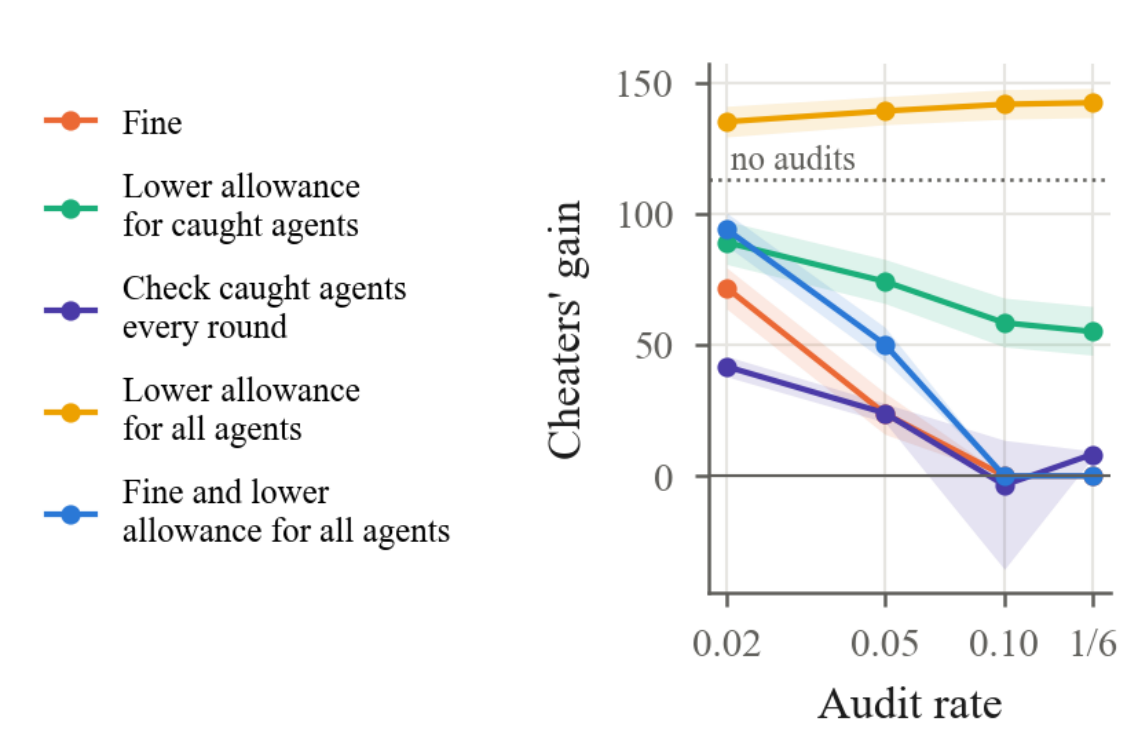
5 of 5 within one step

Every observed threshold landed within one step of the prediction, for thresholds from 0.19 to 1.20.

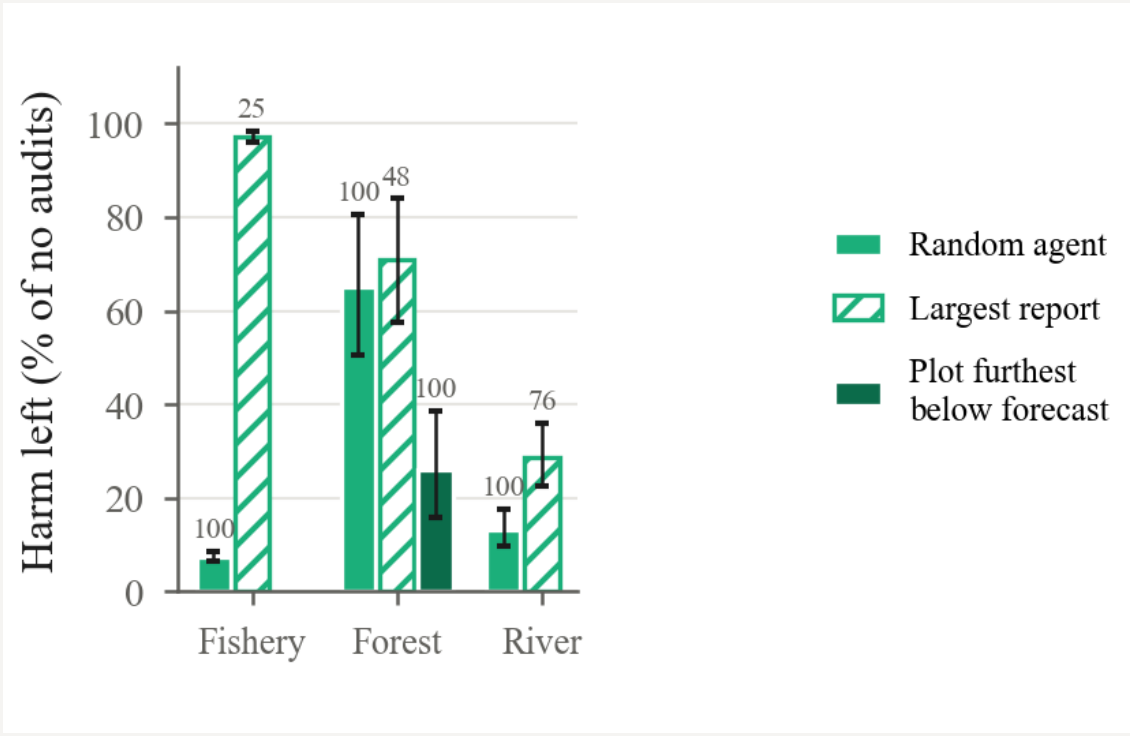
Why it matters

This checks our measurement method before we try it on LLMs.

Finding 3: Two Shortcuts That Backfire



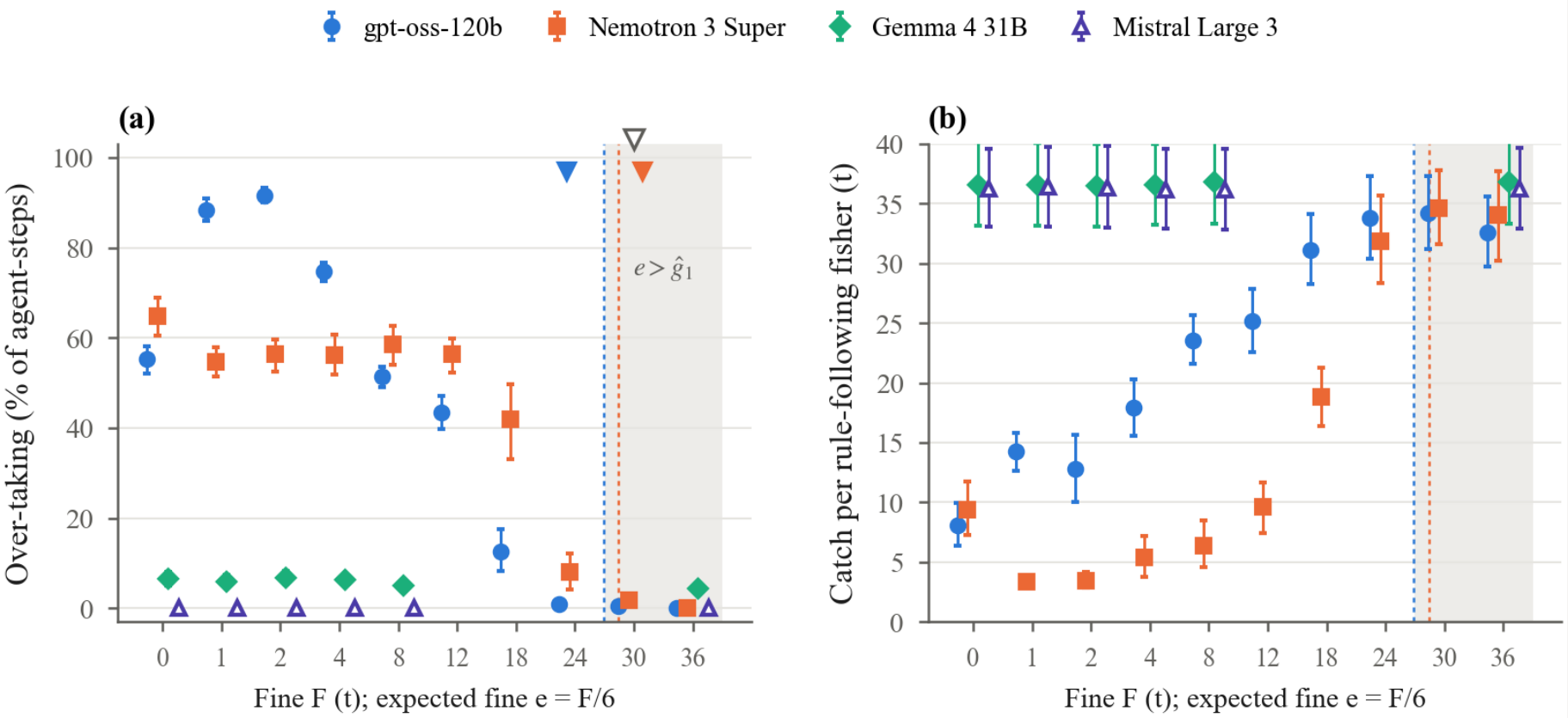
Lowering everyone's allowance to absorb cheaters made cheating pay more than no audits at all (142 vs 113).



Always checking the largest report caught 25% of liars, against 100% with random checks.

Keep checks unpredictable and covering everyone

Finding 4: LLMs Stop Where Predicted



Four LLMs, 520 games, 81,232 decisions. Left: how often each model took more than its allowance, by fine. Triangles: predicted stop (open) and observed stop (filled). Right: fish left for the rule-following agents.

Predicted 30 t: Nemotron stopped at 30 t, gpt-oss one step early at 24 t

Finding 5: They Act on One Round

The gain of a single round

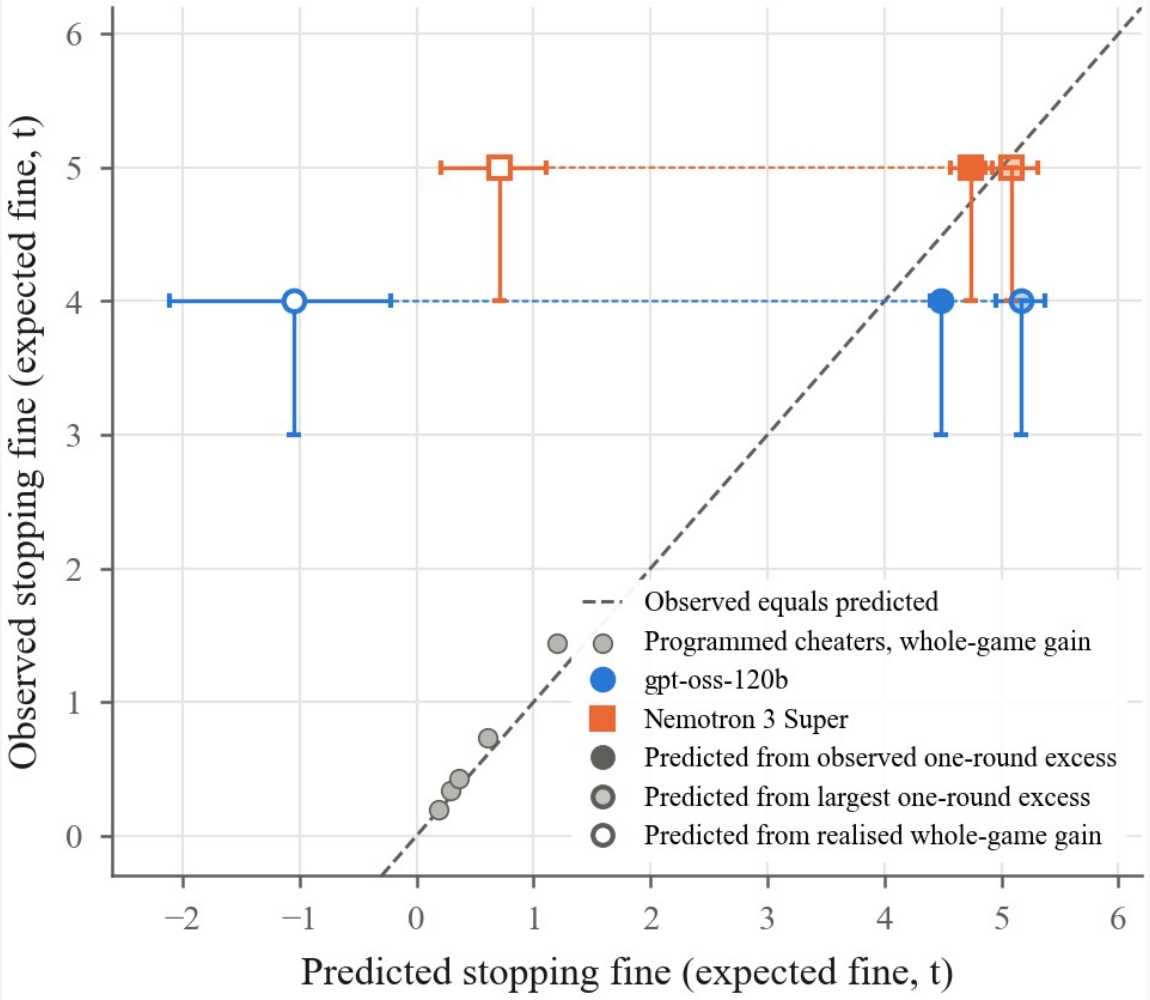
The stopping fine matched what each model took per round without a fine (4.5–4.7 t).

Not the whole game

Over the whole game their over-taking gained little or lost fish (−1.05 and 0.71 t per round), yet they kept going.

Easier to deter than a perfect optimiser

A risk-neutral agent would hold out until 36 t. Both models stopped earlier, and gradually.



Observed against predicted stopping fine. Points on the dashed line are predicted exactly.

Memory Works for LLMs Too

55% → 20%

rounds in which gpt-oss over-took, with memory and no fine

92% → 20%

rounds below half capacity under gpt-oss

0 of 10

lakes collapsed with memory (5 of 10 without)

Reduced, not stopped

Memory protected the resource without any fine, but some over-taking remained. Fines above the gain stopped it and cost no efficiency (0.99 of the maximum sustainable harvest).

Running Now (Results Soon)

Expected fine or fine size?

Same expected fine (2 t or 3 t) with 1 in 3, 1 in 6 or 1 in 12 checked. gpt-oss, 20 new populations.

Result (to fill in):

Does the prediction transfer?

10 new populations and a reworded prompt. Predicted stop frozen in advance: 30 t for both models.

Result (to fill in):

What if nothing checks at all?

Rules that never mention checks, compared with a check that carries a 0 t fine.

Result (to fill in):

Does cheating spread to neighbours?

Forest ring: one programmed over-cutter, five LLM foresters; neighbours' cuts hidden or visible.

Result (to fill in):

Where This Is Going

Sep – 8 Oct

Programmed agents in three games; LLMs in Fishery

12 – 14 Oct

Follow-up tests from two outside reviews

20 Nov

AAAI-27 workshop (WMAC) and arXiv

Jan 2027

Main conference: IJCAI or ICML

Every study has a protocol frozen before its code; failed predictions are reported.

Questions for You

01 Is "taking more than your allowance" a fair stand-in for an AI agent misbehaving? What would be better?

02 If a fine works under one wording of the rules, how far would you trust it under another?

03 What would convince you more: more models, or more kinds of game?

04 Where in your own work does someone check only a sample, and what happens after a catch?

Thank You

13 October 2026

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